

Discrepancies in the Geotargeting of Spanish vs English Google Search Results During the 2024 US Elections

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Abstract

Geotargeting is a common affordance of search engines that enables the tailoring of content based on a user's geolocation. This capability is important during political elections, especially for queries that facilitate participation, such as "where to vote" or "voting near me". While U.S. federal law mandates that certain local election administrators provide this information in multiple languages, such as Spanish, our bilingual audit of Google Search results across 223 U.S. localities reveals significant discrepancies in information quality. We find that geotargeting is often inaccurate at both the county and city levels, with a higher mistargeting rate for Spanish-language results. These findings highlight the need for increased efforts to make important electoral information more discoverable by search engines and accessible through alternative means.

Keywords

geotargeting, search engine audit, bilingual, election information

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1 Introduction

Many people rely on search engines such as Google for information on elections. In 2024, a survey of U.S. voters found that 51% of participants did this.¹ In the same survey, 27% of participants reported seeking out local and county election administrators to find information about voting and registration. Given this information landscape, we understand that when people go to Google with election-related queries, they expect to find the websites of these official government sources as search results. But do the Google search results reliably return official government sources? And does the same hold for languages other than English?

¹<https://bipartisanpolicy.org/article/where-voters-look-for-information-ahead-of-the-2024-election/>



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In the United States, the Voting Rights Act (VRA) Section 203² states that "whenever any State or political subdivision [covered by the section] provides registration or voting notices, forms, instructions, assistance, or other materials or information related to the electoral process, including ballots, it shall provide them in the language of the applicable minority group as well as in the English language". This section of the VRA covers U.S. subdivisions where any language other than English is spoken by more than 5% of the population. The spirit of this provision is to make sure that all U.S. citizens, regardless of their English-proficiency, can meaningfully participate in elections. Given this obligation of election administrators to provide such information in a minority language, and coupling it with the fact that people use Google as their primary resource to access information, we are interested in investigating the extent to which Google surfaces official sources when given an election related query in a language other than English. Since Spanish is the second-most spoken language in the U.S., our focus is on Spanish language results for election information.

Consider the following scenario: a Spanish speaking voter, living in a county where VRA Section 203 applies, makes a Google search in Spanish about election information. Are they able to find information relevant to them? In the context of this analysis, relevance is not only tied to the content of a search, but also to its context, specifically their geographical locality. Concretely, if a voter residing in Cuyahoga County, Ohio, searches "donde votar" (translation: where to vote), what is relevant to them is Spanish information from their own county, and not information about the same query but intended for voters in Clark County, Nevada. This is what we consider as **geotargeting**,³ a form of geolocation-based personalization of search results [7].

Although Google Search is not required by law to comply with VRA Section 203, its outputs still need to be scrutinized by independent entities (such as researchers) to understand barriers to information access encountered by minority voters relying on online platforms. We believe that a comparative audit of search results for English and Spanish queries related to election information is an appropriate tool for this purpose. In the context of such an audit, we ask the following research questions:

- RQ1: To what extent do governmental sources appear for English and Spanish election-related Google Search queries?
- RQ2: What are the differences in geotargeting of results for English and Spanish election-related queries?

Our findings indicate that the search results for the Spanish language surface fewer government web domains than for English, and

²<https://www.justice.gov/crt/language-minority-citizens>

³<https://en.wikipedia.org/wiki/Geotargeting>

even when they do, the geographical mistargetting is significantly higher.

2 Literature Review

Algorithm auditing is the method of repeatedly querying an algorithmic system and observing its output in order to draw conclusions about its inner workings and external impact [13]. This method, rooted in social science methods of testing for racial discrimination [19], has been deployed in a wide-range of contexts over the last fifteen years to study potentially problematic machine behavior [1–3, 12].

Google search in particular has been a platform of particular interest for its widely recognized way of shaping how searchers understand the world [9]. In the context of elections, algorithm audits have focused on different elements of the search results page [6, 8, 11] as well as the ranking order [3].

We are following prior literature such as [10, 14, 16], that takes a voter-centered approach to understanding how people interact with online election information, particularly from search engines. While there has been limited evidence of personalization of search results by user profile, the location that the voter is searching from shapes the quality and content of the search results [5, 16, 17].

Our study contributes to the auditing literature results for a large-scale, bilingual, and geo-diverse audit, which are rare.

3 Data and Methods

To conduct a scraping-audit [18] on geotargeting efficacy, we collected geographic locations, search queries in English and Spanish, and a list of official U.S. government domains. Our materials for this study are available online.⁴

3.1 Selection of Locations

Using 2021 US Census and Department of Justice data [15], we identified 231 localities across 27 US states where Hispanic residents comprise over 5% of the population or exceed 10,000 voting-age citizens with limited English proficiency. Because VRA Section 203 mandates Spanish-language voting materials be provided by election officials in these localities, we hypothesized that search results for election queries would reflect this multi-lingual information availability.

3.2 Selection of Queries

Following prior related literature in voter-centered election audits, we selected 132 election-related English queries from a dataset shared in [14]. These queries were then translated into Spanish by fluent native speakers. The final dataset consists of 264 bilingual queries, covering a range of electoral information categories: ballots, candidates and their issues, general election topics, and voting logistics. Query examples for these categories are shown in Table 1.

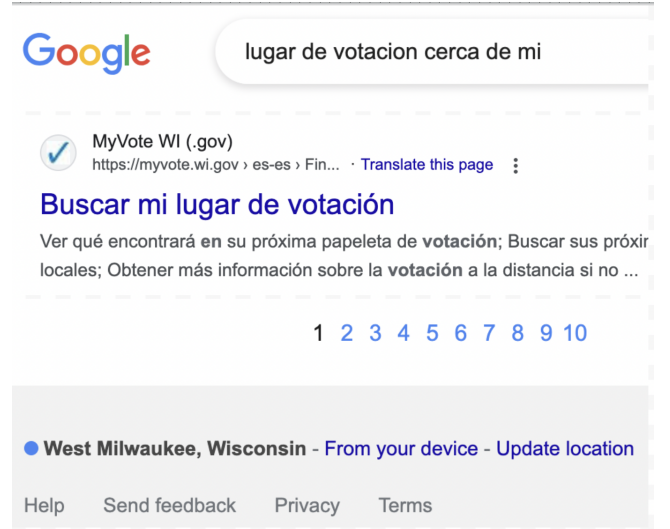


Figure 1: Three elements from a Google Search page: top) Spanish query phrase; middle) the first organic result from an official domain, in Spanish; bottom) the updated web page indicating the locality of the search.

Table 1: Query categories with examples and translations.

Category	English Query	Spanish Translation
Ballots	sample ballot	boleta de muestra
Candidates & Issues	candidate debates	debate de candidatos
General Election Topics	primary election	elecciones primarias
Voting Logistics	election day	dia de elecciones

3.3 SERP Collection

A search engine audit is performed using search queries and geographical locations as inputs, while collecting Search Engine Result Pages (SERPs) as outputs. We ran each of the location-query pairs for English and Spanish simultaneously, collecting a version of the SERP for both languages.⁵

In Figure 1, the sample HTML page of one run for a particular query in Spanish in a particular location in Wisconsin (this screenshot is a stitch image due to space limit). The data collection ran from October 21, 2024 to October 30, 2024, ahead of the November 5, 2024 U.S. Elections. We were able to collect data for 223 of the 231 identified localities and 119 of the 132 queries. Technical issues with Google’s bot detection system limited our ability to collect data for the remaining localities. Overall, we collected 32,690 SERPs with a paired English and Spanish pages, resulting in a final dataset of 65,380 unique SERPs. Each SERP was then decomposed into components: organic results, featured panels, and other page elements resulting in 1,106,096 unique items.

⁴<https://websci26-audit.github.io/>

⁵This process is fully automated. Python scripts simultaneously open two different browsers on the same computer, changing their location to the desired geolocation, submitting the search query, and storing the HTML pages returned by Google.

3.4 Extracting Web Domains

Of the 1,106,096 search components, 794,787 contain URLs. Within these results, there are 16,513 unique links, which belong to 3,769 unique domains overall. (Understandably, there are many domains that have multiple pages related to the elections). The fact that 16,513 links occurred 794,787 times is already a signal that content is not particularly geotargeted, since the same results are showing consistently across our 223 locations. However, our focus is on domains from governmental offices, which we first need to identify.

3.5 Identifying Government Domains

As in related literature [16], we used as the ground truth for government domains a combination of two open source datasets. The U.S. General Services Administration (GSA),⁶ which contains non .gov domains (i.e. .com, .net) and the Cybersecurity and Infrastructure Security Agency (CISA),⁷ which contains .gov domains. These datasets include the location associated with each domain along with its government level. Combining these two lists generates a set of 24,010 domains. Taking the intersection of this set with that of the 3,769 domains extracted from our SERP results, allowed us to find 637 government domains and their respective localities. These domains corresponded to only 56% of the localities in our audit and since this coverage looked low, we discovered that a subset of the official domains in the SERP results were not part of the official datasets. We manually expanded our dataset to include the additional government domains, increasing the number of domains to 926. These cover 76% of the 223 localities in our audit dataset.

3.6 Official Electoral Information in Spanish

Given that official domains from 24% of the localities were missing from our audit results, we manually evaluated that these websites contained election information in Spanish for the corresponding localities. We did this using Google’s domain and phrase search feature to filter each domain for Spanish election phrases such as ‘elección’ and ‘votar’. We found that 77% of these domains contain Spanish election info, confirming that the information exists on these websites.

4 Results

4.1 RQ1: Official Domains in Search Results

The dataset of 926 unique government level domains was represented through 235,726 results in our SERP data, covering 25.70% of all URLs (in both languages). If we break this down by language, this corresponds to 25.70% of all English and 34.67% of all Spanish results in the SERPs belonging to government domains.

Given that these are aggregate values across all locations, queries and domain types, we further break them down to get a more granular view of the distribution of domain appearance and uniqueness of domains for each government level by language. All the results are summarized in Table 2. The English search results involve 858 unique domains while the Spanish language results only include 235 unique domains. Federal domains appear at a higher rate for Spanish queries while state and county-level domains appear at a

higher rate for English queries. As the level of a domain gets more specific (i.e. from federal to city), the percent of their appearance for Spanish queries decreases.

It’s important to point out here that the initial percentages: 25.70% for English and 34.67% for Spanish paint a picture of Spanish results containing more governmental results, which can seem like a positive result. However, the breakdown clarifies that this is because domains at the federal level (e.g. `usa.gov`) are being shown at twice the rate for Spanish, a clear example of lack of geotargeting for showing locality-specific information.

4.2 RQ2: English vs. Spanish Geotargeting

Geotargeting is important because search results which are associated with the **correct** location provide more relevant information. To capture the correctness of geotargeting, we matched the known location of non-federal domain results (that is, domains at the state, county, and city level) to their SERPs’ location. Take `lavote.gov` as an example. This is a county-level domain operating out of Los Angeles County, CA. SERP results which contain this domain are counted as having the correct geotargeting if the SERP query location is Los Angeles County, CA. For state level domains, only the state must be correct. The results of this matching are shown in Table 3. We can notice that official domains are correctly geotargeted at a significantly lower rate for Spanish queries across all non-federal domain levels.

Table 3: Percent correct geotargeting of official domains by level and language.

Level	% Correct Location Eng	% Correct Location Spa
State	74.34	31.64
County	31.19	3.12
City	30.23	7.68

We also investigate the extent to which these results are incorrect. Further analysis of county-level domains finds that when an English county-level result is incorrectly geotargeted, it is still within the correct state 74.34% of the time, compared to just 31.64% of the time for Spanish results. We find similar results for county-level domains, English with 31.19% and Spanish with 3.12%.

Government domains resulting for Spanish queries are more likely to be at the federal level. While websites belonging to federal domains (such as `usa.gov`) provide reliable and abundant information, these websites tend to be less location-specific than state, county and city websites. For locational queries in the Voting Logistics category such as “early voting locations near me”, it is important that non-federal domains appear to provide county and city-specific information. However, the results for these queries match the general trend. Federal domains appear at an overwhelming rate and local results are incorrectly geotargeted.

5 Discussion

5.1 Limited Discoverability for Spanish Content

When we discovered that government domains were not being shown in the SERPs for 24% of the audit localities (in none of

⁶<https://github.com/GSA/govt-urls>

⁷<https://github.com/cisagov/dotgov-data/>

Table 2: The breakdown of search results involving the government domains. Although the percentage of government links for the English language is smaller than those for the Spanish language (25.70% vs. 34.67%), a far larger number of unique domains in English (858) are present compared to only 235 domains for Spanish. This indicates that geotargeted information for Spanish is not easily accessible through search.

Level	English				Spanish			
	Percent	Domain URLs	Uniq. Domains	Queries	Percent	Domain URLs	Uniq. Domains	Queries
Federal	7.50	33,269	31	95	14.95	52,486	20	94
State	10.43	46,276	176	103	11.60	40,731	66	88
County	5.56	24,669	331	84	7.61	26,727	91	81
City	2.21	9,784	320	75	0.51	1,784	58	42
Total	25.70	113,998	858		34.67	121,728	235	

the languages), we manually verified that these domains existed and that the websites also contained Spanish information about elections. This means that the information exists, but that Google Search’s algorithm is not useful in accessing it. When citizens in these jurisdictions search for election information, they are shown some government websites from other voting jurisdictions. While this is a problem for both languages, the disparity between languages is significant. Google is surfacing 3.7 times fewer governmental domains for the Spanish language than for English. Instead, most Spanish queries surface more generic federal-level domains. Google Search’s algorithm is capable of geotargeting, as the English results indicate, but its coverage and accuracy for the Spanish results are subpar. We hope that our research inspires studies for discovering the reasons that could explain this discrepancy.

5.2 Limitations

There are a number of limitations in this study. First, the direct translation of English queries to Spanish may not be entirely representative of how Spanish-speakers search for political information. For example, translating the query “blue wave 2024” gives us “ola azul 2024,” which yielded minimal search results related to politics. Another limitation was the completeness of data collection, which failed on some days due to Google’s reCAPTCHAs. We mitigated this issue by only examining SERPs that had both an English and Spanish counterpart, but we were not able to run through every location-query combination on the same day. Furthermore, our audit is a simulation of geographical location (using Google’s Chrome browser technology), not an authentic event. Google Search algorithms might consider the IP of the request for geotargeting (despite the browser setting) in some fashion that is opaque to us.

5.3 Future Work

All the limitations listed above are a consequence of the experimental set-up of our study (in a lab as opposed to in the field). One could recruit real people, living in various communities around the country and doing their own political searches on their own devices. Such an experiment is in place at the moment through the National Internet Observatory [4], and although it might lack the bi-lingual representation in the recruited sample, it is something to consider for its future expansion.

The electoral information environment is composed of a wide variety of official institutions, news organizations, political groups, individual actors, algorithmic platforms, etc. The domains that we analyzed here only comprise about 1/4 of the domains present across all SERPs (926 out of 3,769). If web domains of official institutions are not being surfaced in the search results, other resources of varying reliability might be filling their place. Being able to categorize the nature and local affiliation of the other domains would contribute to a better understanding of the electoral online information environment.

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